

Week 3 Assignment

Databases Behind APIs: Two Real-World Project Tracks

Choose **ONE** of the following **tracks** to do for week3 assignment and **submit** your code using the **Submission Instructions** at the **end of this document**:

- Track A: Ateko (Commerce and Product Platform)
- Track B: RBC (Banking and Account Platform)

Both tracks require:

- FastAPI + SQLAlchemy ORM
 - MySQL database
 - Postman collection
 - Swagger working
 - Pagination (offset and cursor)
 - Connection pooling
 - One endpoint implemented twice (ORM and raw SQL)
-

1 Deliverables (Both Tracks)

1. Git repository link
 2. Source code (FastAPI + ORM models)
 3. SQL script or migration notes to create tables
 4. Postman collection export
 5. Screenshot of Swagger endpoints showing they work
 6. Short answers document (**Part3 Answers.txt**, **Part4 Answers.txt**)
-

2 Grading Rubric

- ORM models: 25%
- API correctness and validation: 25%
- Pagination (offset and cursor): 20%
- Connection pooling configuration and explanation: 10%
- ORM vs raw SQL analytics and reflection: 20%

What I will look for

- Clean filtering queries
- Correct pagination behavior
- Correct join and aggregation in analytics

3 Submission Instructions

You must submit your work as a **public GitHub repository**.

3.1 Create a Git Repository

Create a repository named:

- `Week3_Ateko_Backend`
or
- `Week3_RBC_Backend`

depending on your track.

3.2 Required Files in Your Repository

Your repository must include:

- All `.py` source files
- The database `.sql` file
- `requirements.txt`
- `README.md`
- `.env.example` (do NOT upload your real `.env`)
- `Part3_Answers.txt`
- `Part4_Answers.txt`

Make sure your project runs successfully.

3.3 README Must Include

Brief instructions on:

- How to import the database
- How to install dependencies
- How to run the FastAPI app

Keep it clear and simple.

3.4 Submit on Brightspace

1. Make your repository **Public**
2. Copy the repository link
3. Submit the link under **Week 3 Submission** on Brightspace

No zip files.

No email submissions.

Before submitting, double-check that:

- The repository is public
- All required files are uploaded
- The link works